



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 14																									
Name of Student	Aryan Sheladia																								
PRN	23070122202																								
Batch	2023 - 2027																								
Class	CSE – B1																								
Academic Year & Semester	2026 - 2027, 7th Semester																								
Title of Assignment	CI/CD – Jenkins Pipeline building Docker Image																								
Implementation & Theory	<pre>PS E:\DevOps_SEM_7> docker ps</pre><table><tr><th>CONTAINER ID</th><th>IMAGE</th><th>COMMAND</th><th>CREATED</th><th>STATUS</th><th>PORTS</th></tr><tr><td>19e57692e73a</td><td>jenkins/jenkins:lts-jdk17</td><td>"/usr/bin/tini -- /u..."</td><td>10 seconds ago</td><td>Up 9 seconds</td><td>0.0.0.0:8080->8080</td></tr></table><pre>00->50000/tcp myjenkins PS E:\DevOps_SEM_7></pre> Jenkins Container with Docker Access: The existing Jenkins container was recreated while preserving the jenkins_home Docker volume. The Docker socket was mounted inside the Jenkins container using /var/run/docker.sock:/var/run/docker.sock, allowing Jenkins to communicate with the host Docker daemon for building Docker images. <pre>root@19e57692e73a:/# docker --version Docker version 26.1.5+dfsg1, build a72d7cd root@19e57692e73a:/# docker ps</pre><table><tr><th>CONTAINER ID</th><th>IMAGE</th><th>COMMAND</th><th>CREATED</th><th>STATUS</th><th>PORTS</th></tr><tr><td>19e57692e73a</td><td>jenkins/jenkins:lts-jdk17</td><td>"/usr/bin/tini -- /u..."</td><td>2 minutes ago</td><td>Up 2 minutes</td><td>0.0.0.0:8080->8080/tcp, :::8080</td></tr></table><pre>root@19e57692e73a:/# exit exit</pre> Docker CLI Installed in Jenkins: Docker was installed inside the Jenkins container and verified using the Docker CLI. The successful execution of docker ps confirms that Jenkins can communicate with the Docker daemon through the mounted Docker socket, enabling Jenkins pipelines to build Docker images.	CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	19e57692e73a	jenkins/jenkins:lts-jdk17	"/usr/bin/tini -- /u..."	10 seconds ago	Up 9 seconds	0.0.0.0:8080->8080	CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	19e57692e73a	jenkins/jenkins:lts-jdk17	"/usr/bin/tini -- /u..."	2 minutes ago	Up 2 minutes	0.0.0.0:8080->8080/tcp, :::8080
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```

36     }
37
38     stage('Build Docker Image') {
39         steps {
40             dir('Assignment - 8') {
41                 sh 'docker build -t my-flask-app:${BUILD_NUMBER} .'
42             }
43         }
44     }
45 }
46 }

```

Updated Jenkins Pipeline: The existing Declarative Jenkins Pipeline was modified to include a new **Build Docker Image** stage. This stage navigates to the Assignment - 8 directory containing the Flask application's Dockerfile and builds a Docker image. The `${BUILD_NUMBER}` variable automatically assigns a unique tag to each image generated by Jenkins.

 #5 (Sep 4, 2026, 8:32:55 AM)



Started by user [Aryan Sheladia](#)



Revision: 73a38f0760a52805497c231f78f74ad63a2320f3

Repository: https://github.com/AryanSheladia/DevOps_SEM_7.git

- refs/remotes/origin/main



No changes.

Successful CI/CD Pipeline Execution: The Jenkins Declarative Pipeline executed successfully. All stages completed without errors, including source code checkout, parameterized greeting, command execution, and Docker image building.

```

[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] dir
Running in /var/jenkins_home/workspace/Declarative-Pipeline-Demo/Assignment - 8
[Pipeline] {
[Pipeline] sh
+ docker build -t my-flask-app:5 .
#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 204B done
#1 DONE 0.0s

#2 [internal] load metadata for docker.io/library/python:3.9-slim
#2 DONE 1.5s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

```

Docker Image Built by Jenkins: The Build Docker Image stage successfully executed the Docker build command inside Jenkins. The Flask application image was built from the Dockerfile in Assignment 8 and tagged automatically using the Jenkins build number.

```

PS E:\DevOps_SEM_7> docker images

```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
my-flask-app	5	e42d438f7aa6	7 minutes ago	200MB
gcr.io/k8s-minikube/kicbase	v0.0.51	146cd636030e	2 days ago	1.89GB

Docker Image Verification: The docker images command confirmed that the Docker image was successfully created by the Jenkins Pipeline. The image is tagged with the Jenkins build number, providing a unique version for each pipeline execution.